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Centre Number

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Student Number

2020

Year 12

Mathematics Advanced

Task 3

Monday, 25th May

**General
Instructions:**

- Reading time – 50 minutes
- Working time – 5 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:
44

Q	Marks
1	11
2	6
3	9
4	3
5	15
Total	/44

This question paper must not be removed from the examination room.

This assessment task constitutes 25% of the course.

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Question 1 (11 marks)

(a) Differentiate the following:

(i) $f(x) = x^3 - e^x$ 1

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.....

(ii) $f(x) = x\sqrt{x}$ 2

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(ii) $f(x) = \tan(x^2)$ 2

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(iii) $y = \frac{\sin x}{x}$ 2

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(iv) $y = \log_3 x$

1

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(v) $f(x) = \ln(e^x(x + 2))$

3

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Question 2 (6 marks)

- (a) Given the curve $y = x^4 - 8x^2$.
Find the stationary points and determine their nature.

3

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(b) Determine the point of inflection of the curve $y = xe^x$.

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Question 3 (9 marks)

(a) Find the following.

(i) $\int 3 - x \, dx$

2

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(ii) $\int \frac{2}{x} dx$

1

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(iii) $\int_0^4 (x-1)^3 dx$

2

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(iv) $\int_{-2}^0 e^{3x} dx$

2

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(v)

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{2} - \cos x \, dx$$

2

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Question 4 (3 marks)

(a) (i) Differentiate $f(x) = \ln(x^2 + 16)$.

1

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(ii) Hence or otherwise, find $\int_0^3 \frac{x}{x^2+16} \, dx$.

2

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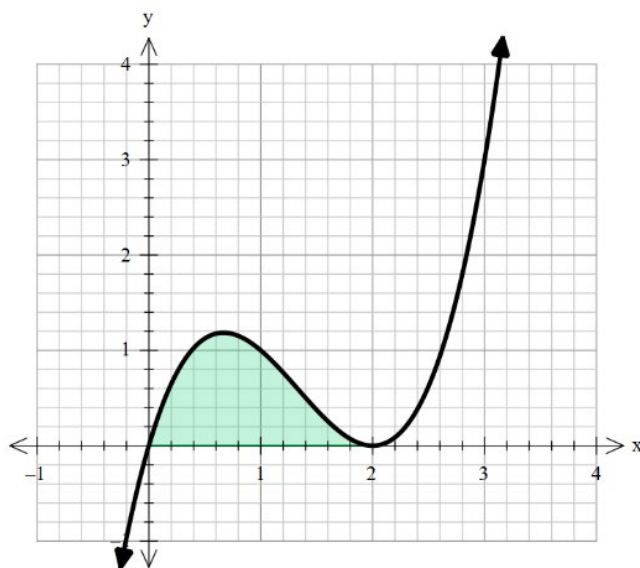
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Question 5 (15 marks)

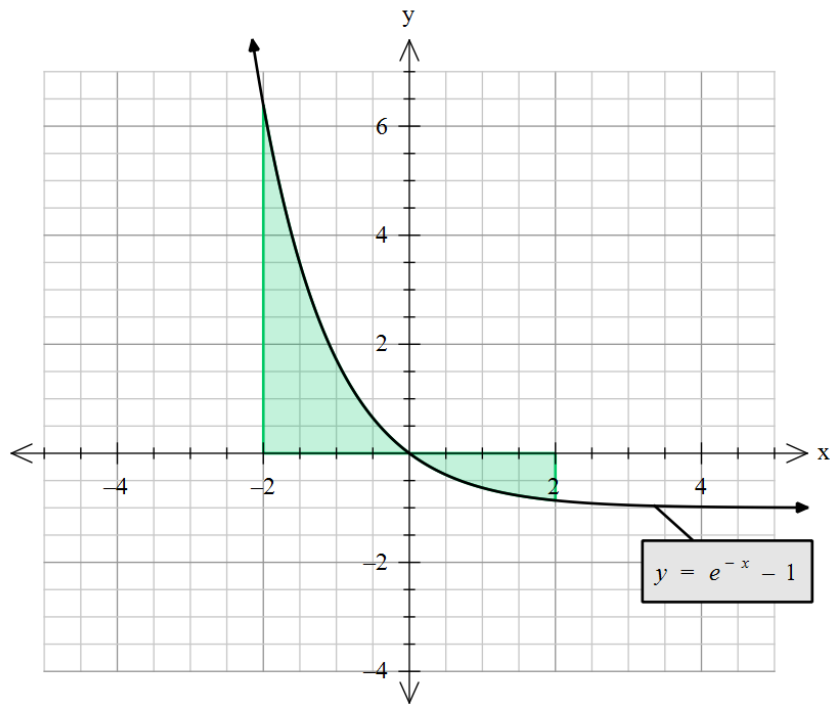
- (a) Find the exact area of the region bounded by the curve $y = x^3 - 4x^2 + 4x$.

2

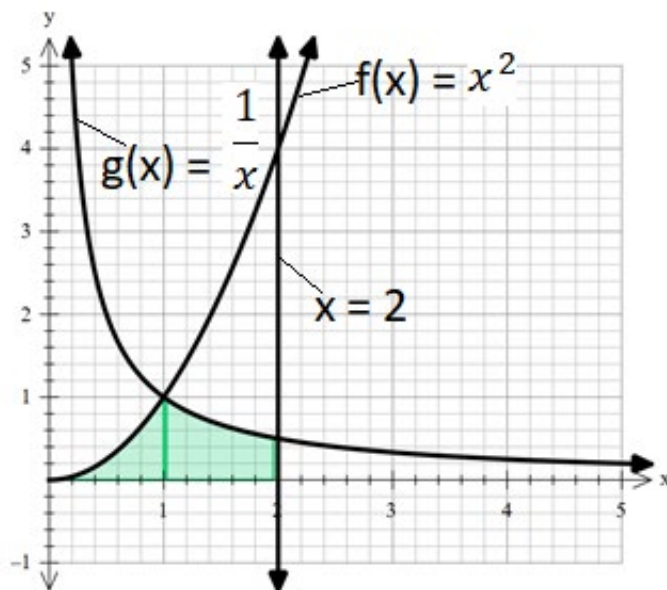
This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (b) Find the exact area of the region bounded by the curve $y = e^{-x} - 1$.

2

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (c) Calculate the exact area bounded by $f(x) = x^2$, $g(x) = \frac{1}{x}$, $x > 0$, the x-axis and the line $x = 2$.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(d) (i) Complete the table of values.

1

x	1	2	3	4
$\frac{1}{x^2 + 1}$				

(ii) Approximate $\int_1^4 \frac{1}{x^2 + 1} dx$ using the trapezoidal rule with 3 sub-intervals.
Answer in 2 decimal places.

2

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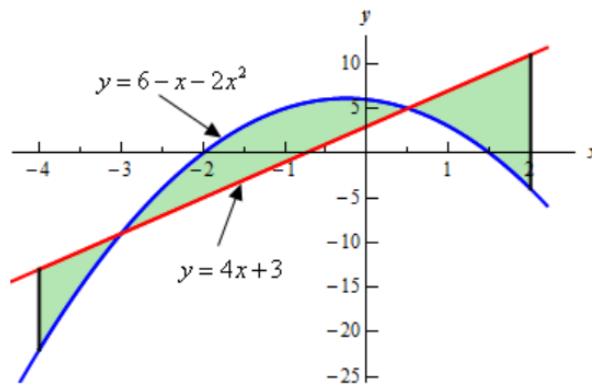
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- (e) Determine the exact area of the region bounded by $y = 4x + 3$, $y = 6 - x - 2x^2$, $x = -4$ and $x = 2$. 5

[illegible]

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.



Killara
HIGH SCHOOL

MATHEMATICS DEPARTMENT

Solutions with Marking Scheme

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Question 1 (11 marks)

(a) Differentiate the following:

(i) $f(x) = x^3 - e^x$

1

$$f'(x) = 3x^2 - e^x \quad \checkmark$$

(ii) $f(x) = x\sqrt{x}$

2

$$\begin{aligned} f(x) &= x^{\frac{3}{2}} \quad \checkmark \\ f'(x) &= \frac{3}{2} x^{\frac{1}{2}} \\ &= \frac{3\sqrt{x}}{2} \quad \checkmark \end{aligned}$$

(ii) $f(x) = \tan(x^2)$

2

$$f'(x) = \underbrace{2x}_{\checkmark} \underbrace{\sec^2(x^2)}_{\checkmark}$$

1 mistake;
1 mark off.

(iii) $y = \frac{\sin x}{x}$

2

$$\begin{aligned} u &= \sin x & v &= x \\ u' &= \cos x & v' &= 1 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \checkmark$$

$$y' = \frac{u'v - uv'}{v^2}$$

$$= \frac{x \cos x - \sin x}{x^2} \quad \checkmark$$

1 mistake;
1 mark off

(iv) $y = \log_3 x$

1

$$y' = \frac{1}{\log_e 3 \times x} \quad \checkmark$$

(v) $f(x) = \ln(e^x(x+2))$

3

$$f(x) = \ln e^x + \ln(x+2) \quad \checkmark$$

1 mistake;

$$f'(x) = x + \ln(x+2) \quad \checkmark$$

1 mark off.

$$f'(x) = 1 + \frac{1}{x+2} \quad \checkmark$$

$$= \frac{x+2+1}{x+2}$$

$$= \frac{x+3}{x+2} \quad \checkmark$$

Question 2 (6 marks)

(a) Given the curve $y = x^4 - 8x^2$.

3

Find the stationary points and determine their nature.

1 mistake;
1 mark off

$$y' = 4x^3 - 16x$$

$$y' = 0, \quad 4x(x^2 - 4) = 0$$

$\therefore$ Stationary points at $x = \pm 2, x = 0$

$$y'' = 12x^2 - 16$$

$$y''(0) = -16 < 0$$

$$y''(2) = 32 > 0, \quad y''(-2) = 32 > 0$$

$\therefore$ Maximum point at $x = 0$

minimum points at $x = -2$ and $x = +2$

- (b) Determine the point of inflection of the curve $y = xe^x$.

3

1 mistake!
1 mark of

$$\begin{aligned} y' &= e^x + xe^x \\ y'' &= e^x + e^x + xe^x \\ &= 2e^x + xe^x \end{aligned}$$

$$\begin{aligned} y'' &= 0. & e^x(2+x) &= 0 \\ \therefore e^x &\neq 0. & 2+x &= 0. \\ \therefore & \text{when } x = -2 \end{aligned}$$

$\therefore$ The point of inflection at $x = -2$.

x	-3	-2	-1
y''	-	0	+
Concavity	$\cap$	.	$\cup$

$$y(-2) = -2e^{-2} = \frac{-2}{e^2}$$

$\therefore$ The point inflection $(-2, \frac{-2}{e^2})$

Question 3 (9 marks)

- (a) Find the following.

(i) $\int 3 - x \, dx$

2

$$= 3x - \frac{x^2}{2} + C$$

1 mark off
not
missing +C

(ii) $\int \frac{2}{x} dx$

1

$$= 2 \ln x + C \checkmark$$

no mark
taking off
for not having
"+C".

(iii) $\int_0^4 (x-1)^3 dx$

2

$$= \left[\frac{(x-1)^4}{4} \right]_0^4 \checkmark$$

$$= \frac{81}{4} - \frac{(-1)^4}{4}$$

$$= \frac{80}{4} = 20 \checkmark$$

(iv) $\int_{-2}^0 e^{3x} dx$

2

$$= \left[\frac{e^{3x}}{3} \right]_{-2}^0 \checkmark$$

$$= \frac{1}{3} - \frac{e^{-6}}{3}$$

$$= \frac{1 - e^{-6}}{3} \checkmark$$

$$= \frac{e^6 - 1}{3e^6} \checkmark$$

$$\approx 0.3325 \checkmark$$

(v)

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{2} - \cos x \, dx$$

2

$$= \left[-\frac{\cos x}{2} - \sin x \right]_0^{\frac{\pi}{2}} \checkmark$$

$$= - \left[\frac{\cos x}{2} + \sin x \right]_0^{\frac{\pi}{2}}$$

$$= - \left[(0 + 1) - \left(\frac{1}{2} - 0 \right) \right]$$

$$= -\frac{1}{2} \checkmark$$

Question 4 (3 marks)

(a) Evaluate the following:

- (i) ^{Differentiate} Find the derivative of $f(x) = \ln(x^2 + 16)$.

1

$$f'(x) = \frac{2x}{x^2 + 16} \checkmark$$

- (ii) Hence or otherwise, find $\int_0^3 \frac{x}{x^2 + 16} \, dx$.

2

$$= \left[\frac{\ln(x^2 + 16)}{2} \right]_0^3 \checkmark$$

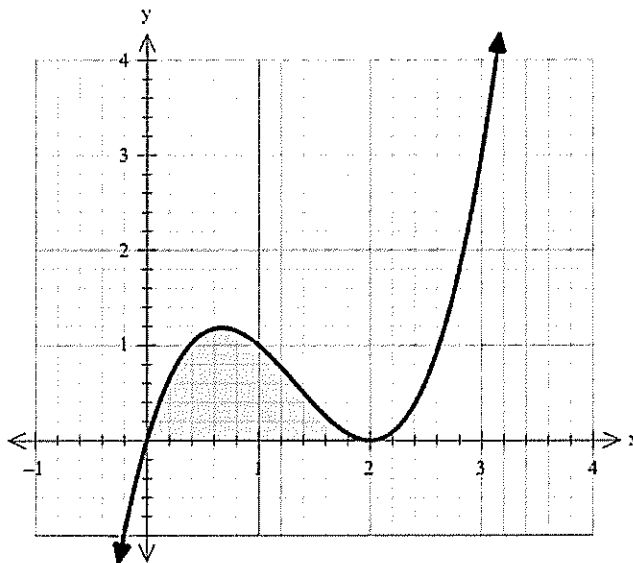
$$= \frac{\ln 5^2}{2} - \frac{\ln 4^2}{2} = \ln 5 - \ln 4 \approx 0.22 \checkmark$$

OR $\checkmark$

Question 5 (15 marks)

- (a) Find the ^{exact} area of the region bounded by the curve $y = x^3 - 4x^2 + 4x$.

2



$$\int_0^2 x^3 - 4x^2 + 4x \, dx$$

$$= \left[\frac{x^4}{4} - \frac{4x^3}{3} + 2x^2 \right]_0^2 \quad \checkmark$$

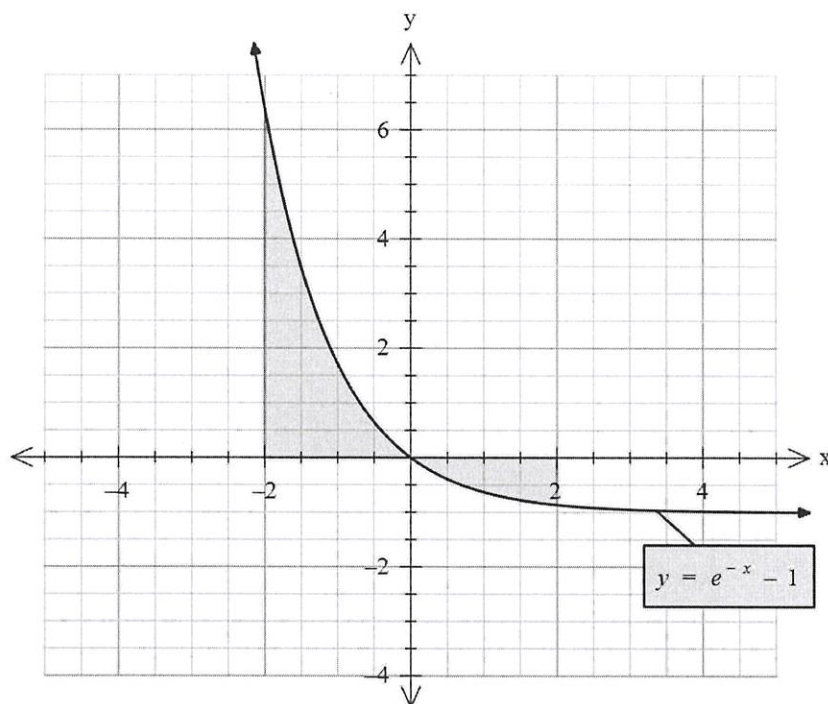
$$= \left[\frac{16}{4} - \frac{32}{3} + 8 \right] - 0$$

$$= 12 - \frac{32}{3}$$

$$= \frac{4}{3} \quad \checkmark$$

(b) Find the exact area of the region bounded by the curve $y = e^{-x} - 1$.

2



1 mark off:
1 mistake

$$\int_{-2}^2 e^{-x} - 1 \, dx$$

$$= \int_{-2}^0 e^{-x} - 1 \, dx + \left| \int_0^2 e^{-x} - 1 \, dx \right| \checkmark$$

$$= [-e^{-x} - x]_{-2}^0 + |[-e^{-x} - x]_0^2|$$

$$= -[e^{-x} + x]_{-2}^0 + [e^{-x} + x]_0^2$$

$$= -(1 - (e^2 - 2)) + ((e^{-2} + 2) - (1))$$

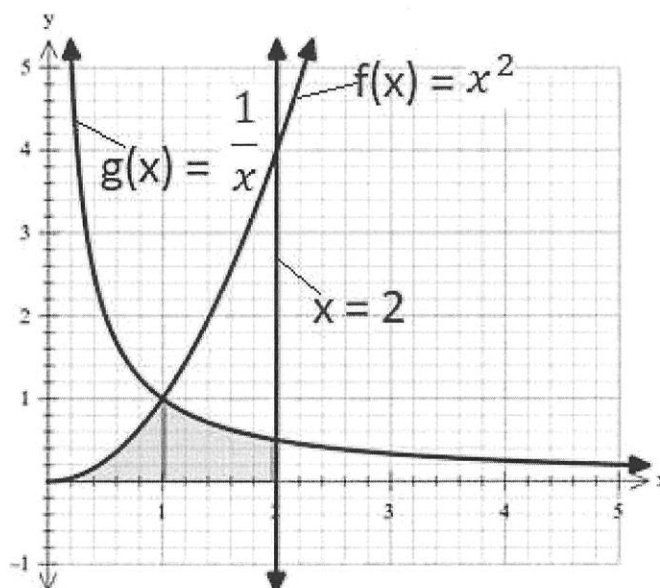
$$= -(1 - e^2 + 2) + (e^{-2} + 1)$$

$$= e^2 - 3 + e^{-2} + 1$$

$$= -2 + e^2 + e^{-2} \checkmark$$

- (c) Calculate the exact area bounded by $f(x) = x^2$, $g(x) = \frac{1}{x}$, $x > 0$, the x-axis and the line $x = 2$.

3



Total area

$$= \int_0^1 x^2 dx + \int_1^2 \frac{1}{x} dx \quad \checkmark$$

$$= \left[\frac{x^3}{3} \right]_0^1 + [\log_e x]_1^2 \quad \checkmark$$

$$= \frac{1}{3} + (\ln 2 - \ln 1)$$

$$= \frac{1}{3} + \ln 2$$

$$= \frac{1}{3} + \log_e 2 \quad \text{OR} \quad \checkmark$$

$$= \frac{1 + 3 \log_e 2}{3} \quad \text{OR} \quad \checkmark$$

1 mark off;
1 mistake.

(d) (i) Complete the table of values.

1

x	1	2	3	4
$\frac{1}{x^2+1}$	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{17}$ ✓

(ii) Approximate $\int_1^4 \frac{1}{x^2+1} dx$ using the trapezoidal rule with 3 sub-intervals. Answer ²
in 2 decimal places.

$$\int_1^4 \frac{1}{x^2+1} dx$$

$$\approx \frac{1}{2} \times [f(1) + 2f(2) + 2f(3) + f(4)] \checkmark$$

$$\approx \frac{1}{2} \times \left[\frac{1}{2} + \frac{2}{5} + \frac{2}{10} + \frac{1}{17} \right] \checkmark \text{ OR}$$

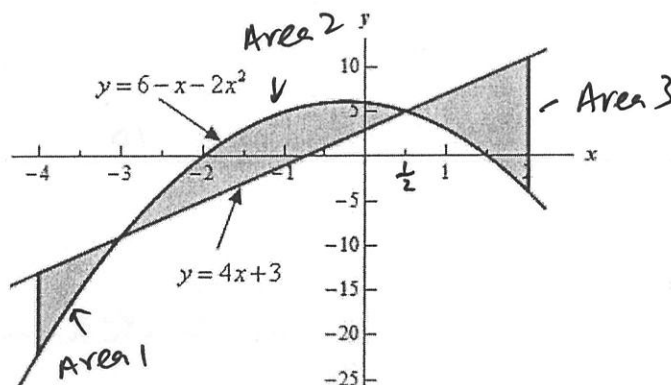
$$= \frac{1}{2} \times \frac{197}{170}$$

$$\approx \frac{197}{340} \checkmark$$

exact

- (e) Determine the area of the region bounded by $y = 4x + 3$, $y = 6 - x - 2x^2$, $x = -4$ and $x = 2$.

5



Total Area 1

$$= \int_{-4}^3 (4x+3 - (6-x-2x^2)) dx$$

$$= \int_{-4}^3 (4x+3-6+x+2x^2) dx$$

$$= \int_{-4}^3 (2x^2+5x-3) dx$$

$$= \left[\frac{2x^3}{3} + \frac{5x^2}{2} - 3x \right]_{-4}^3$$

$$= \left(-18 + \frac{45}{2} + 9 \right) - \left(\frac{2(-4)^3}{3} + 40 + 12 \right)$$

$$= \frac{27}{2} - \frac{28}{3} = \frac{25}{6}$$

Area 2

$$= \int_{-3}^{0.5} (6-x-2x^2 - 4x-3) dx$$

$$= \int_{-3}^{0.5} (-2x^2-5x+3) dx$$

$$= \left[-\frac{2x^3}{3} - \frac{5x^2}{2} + 3x \right]_{-3}^{0.5}$$

$$= \left(-\frac{1}{12} - \frac{5}{8} + 1.5 \right) - \left(\frac{1}{3} 18 - \frac{45}{2} - 9 \right)$$

$$= \frac{343}{24}$$

Area 3

$$= \int_{\frac{1}{2}}^2 (2x^2+5x-3) dx = \left[\frac{2x^3}{3} + \frac{5x^2}{2} - 3x \right]_{\frac{1}{2}}^2$$

$$= \left(\frac{16}{3} + 10 - 6 \right) - \left(\frac{1}{12} + \frac{5}{8} - \frac{3}{2} \right)$$

$$= \frac{81}{8}$$

$$\therefore \text{Total area} = \frac{25}{6} + \frac{343}{24} + \frac{81}{8} = \frac{343}{12}$$

each
- 1 mark for
correct definite
integral of each
area.

- 1 mark for
correct working

- 1 mark for
correct answer.